

Recursion + Induction

claim $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for all $n \geq 1$.

ex $n=5$ $\sum_{i=1}^5 i = \underline{1} + \underline{2} + \underline{3} + \underline{4} + \underline{5} = 15$

LHS = 15

$$\frac{5(6)}{2} = \frac{30}{2} = 15$$

RHS = 15

for $i=0, 1, \dots, n$:

for $j=i+1, i+2, \dots, n$:

do something

$\sum_{i=1}^0 i = 0 = \text{LHS}$
 $\frac{0(0+1)}{2} = 0$ RHS

claim $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for all $n \geq 0$.

proof We prove the claim by induction on n .

let n be an arbitrary integer ≥ 1 .

Assume that for all $n' < n$,

$$\sum_{i=1}^{n'} i = \frac{n'(n'+1)}{2}$$

Induction hypothesis, IH

We prove by cases.

Suppose $n=0$. Then $LHS = \sum_{i=1}^0 i = 0$.

$$RHS = \frac{0(0+1)}{2} = \frac{0}{2} = 0. \checkmark$$

Suppose $n=1$

Suppose $n > 1$. Then

$$LHS = \sum_{i=1}^n i$$

$$\sum_{i=1}^{n-1} i$$

$$= 1 + 2 + \dots + n-1 + n$$

$$\frac{(n-1)(n-1+1)}{2}$$

$$= \left(\sum_{i=1}^{n-1} i \right) + n$$

$$= \frac{(n-1)n}{2}$$

$$= \frac{(n-1)n}{2} + n$$

by IH

$$= \frac{n^2 - n}{2} + n$$

$$= \frac{n^2 - n}{2} + \frac{2n}{2}$$

$$= \frac{n^2 - n + 2n}{2}$$

$$= \frac{n^2 - n}{2}$$

$$= \frac{n(n+1)}{2}$$

$$= RHS$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2} \leftarrow$$

LHS = RHS

Since the claim holds in all cases, and the cases are exhaustive, $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for all $n \geq 1$.

Proof commandment # : Thou shalt not assume the claim!

~~ex~~ claim $n = O(n^2)$. ~~X~~
proof let $n = O(n^2)$.

n	1	2	3	...
claim T?	Y	Y	Y	
why?	direct proof	we showed if claim T $\forall n' < n$ then claim T for n and claim T for $n=1$	we showed that if claim T $\forall n' < n$ then claim T for n and claim T for $n=1$ and $n=2$	

"principle of mathematical induction"

Proof by Induction Boilerplate

Claim All X have property Y

$$\left(\begin{array}{l} \forall n \geq 1 \\ \text{have} \\ \sum = \frac{n(n+1)}{2} \end{array} \right)$$

Proof let z be an arbitrary x .

① Universal declaration

Assume that all z' smaller than z have property Y .

② Induction hypothesis

cases

③ Base case: don't use IH

④ Inductive case: use IH

⑤ wrap-up sentence